SIMATS ENGINEERING

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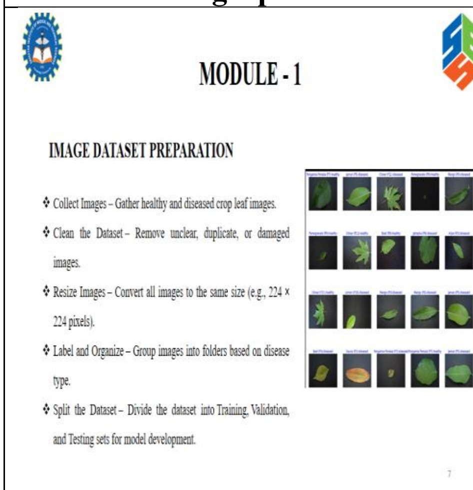
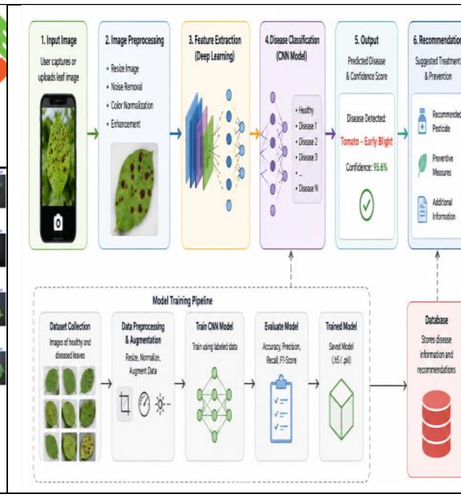
Course Code: CSA 1711 **Slot:** C

Course Name: Artificial intelligence

Course Faculty: Dr. K. Krishna Kumar

Project Title: AI - Based crop disease detection
Module Photographs:





Project Description: MODULE 1 : IMAGE DATASET PREPARATION

The Image Dataset Preparation module is responsible for collecting and preparing plant leaf images from the PlantVillage dataset available on Kaggle. The images are organized according to different crops and disease classes, then cleaned and resized to a fixed size suitable for the CNN model. The dataset is divided into 70% training, 15% validation, and 15% testing data to ensure proper model training and evaluation. Data augmentation techniques such as flipping, rotation, and zooming can also be applied to increase image variety and improve the model's ability to recognize plant diseases accurately.

The images are converted into a suitable format and normalized before being given to the model. Each disease category is assigned a unique class label automatically from its folder name. This module ensures that the dataset is properly structured and ready for CNN training. It also helps reduce errors and prevents overlap between training, validation, and testing images. Finally, the prepared dataset acts as the main input for the disease classification model.

Student Signature **Guide Signature**